

Benefits & Risks Evaluation Modeling

Dynamic evaluation of a drug's
potential, assisted by AI





Facts & Issues

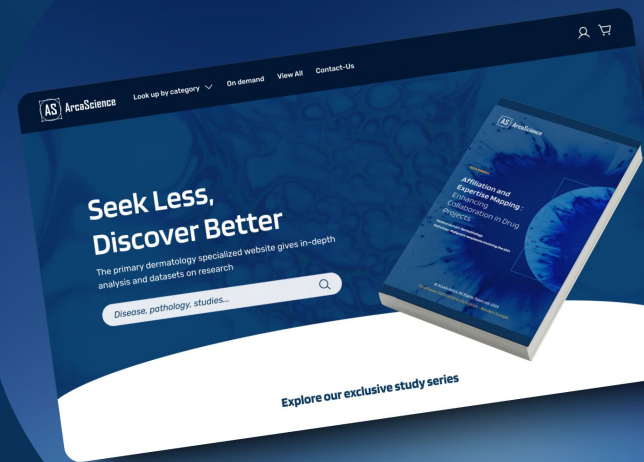
- +10 years to bring a drug to market
- Fails 9/10 times & costs an average of \$2.3B per drug
- Reasons:
 - Lack of efficacy 57%
 - Safety issues 17%

Sources: Annexes

Vision

Every strategic clinical decision is based on a benefit-risk assessment.

A **dynamic** benefit-risk assessment
is the key to effective medical decision-making.



ArcaScience Dramatically Changes the way these Evaluations are Conducted

Build your drugs' BRE in seconds

Don't spend months looking for it

By far the most
comprehensive BRE
database in the world

+

24 small AI models
trained by clinicians

Unmatched, with over 100bn data points

...



Delivered on a **user-friendly & transparent** solution dashboard

Benefits & Risks
strengths, weaknesses &
opportunities



Approval **optimization**
*Risk mitigation, new design
amendment strategies,
better applications*

Reduce BRA Project Time & Study Generation by 80%¹

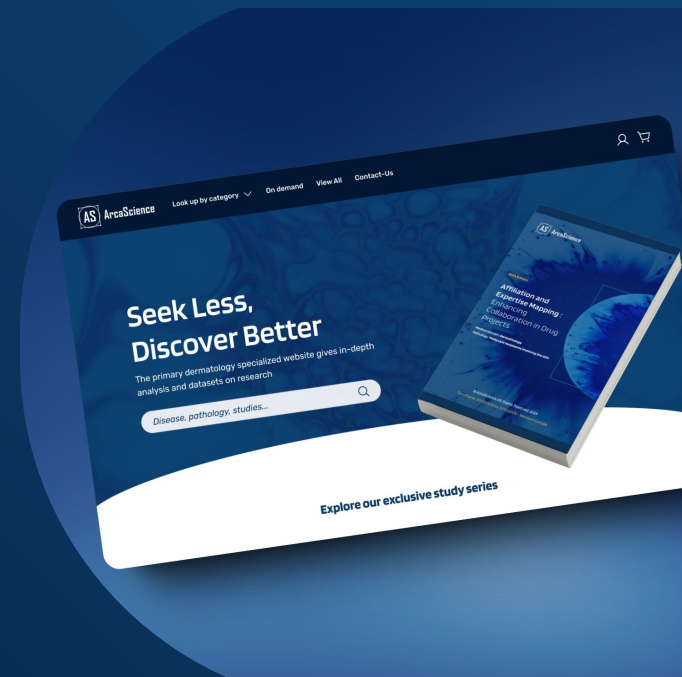
¹: independent BRA experts evaluation, 2025

Our vision

Making sure drugs remain best-in-class

From the drug candidate
selection & acquisition phase

All the way down to the market approval



Current Obstacles in Pharmaceutical Benefit-Risk Evaluation

1

Need for
comprehensive
integration of
medical evidence

2

Overly long
analysis duration

3

Need to increase
characterisation
in risk assessment

4

Challenges in
real-time
evidence
integration

5

Insufficient
integration of
patient
preferences

Innovative Approaches are Transforming Benefit-Risk Analysis

Automated Studies Generation that Fit CIOMS Framework

Product Opportunities

- ✓ Analysis of disease/condition
- ✓ Current treatment options

Product Profile

- ✓ Key Benefits & Risks
- ✓ Uncertainties for Benefits & Risks

Product Optimisation

Risk Management
(in progress)

✓ **Benefit-Risk Conclusion**

Trusted by Leading Pharma Groups:

AstraZeneca

sanofi

Takeda

✓ : Partial coverage

✓ : Available studies

Six Pre-Configured Study Types generated by our AI platform

Analysis of condition



Disease Analysis

Discover molecular pathways and epidemiological information about your disease

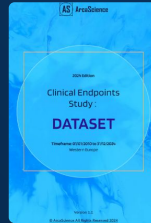
Current Treatments



Clinical Landscape + Efficacy Report

Access to a global synthesis of clinical activities with sponsors, treatments...

Benefits



Clinical Endpoint Study

Access to a accurate analysis of all clinical endpoint strategy for a pathology

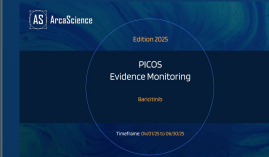
Risks



Adverse Events Reports

Get a synthesis of all adverses events related to your drug and the standard of care

Benefit-Risk Assessment



Benefit Risk Assessment

Access to a summarized BRA regarding your priorities

Conclusion



Benefit Risk Summary

Review a synthesis of your tailored benefit-risk analysis

Each Study Delivers Full Data Transparency and FDA-Compliant Documentation

Studies Alone Falls Short in Two Critical Areas

Agility in the BRA Process (Test & Learn)



Real-Time Updates

Our BRA Platform Manages Safety Team Priorities & Workflow

BRAT WORKFLOW

- ✓ Step 0: Configuration
- ✓ Step 1: Context
- ✓ Step 2: Outcomes Identification
- ✓ Step 3: Data Analysis
- ✓ **Step 4: Customize**
- ✓ Step 5: Weighting
- ✓ Step 6: Display

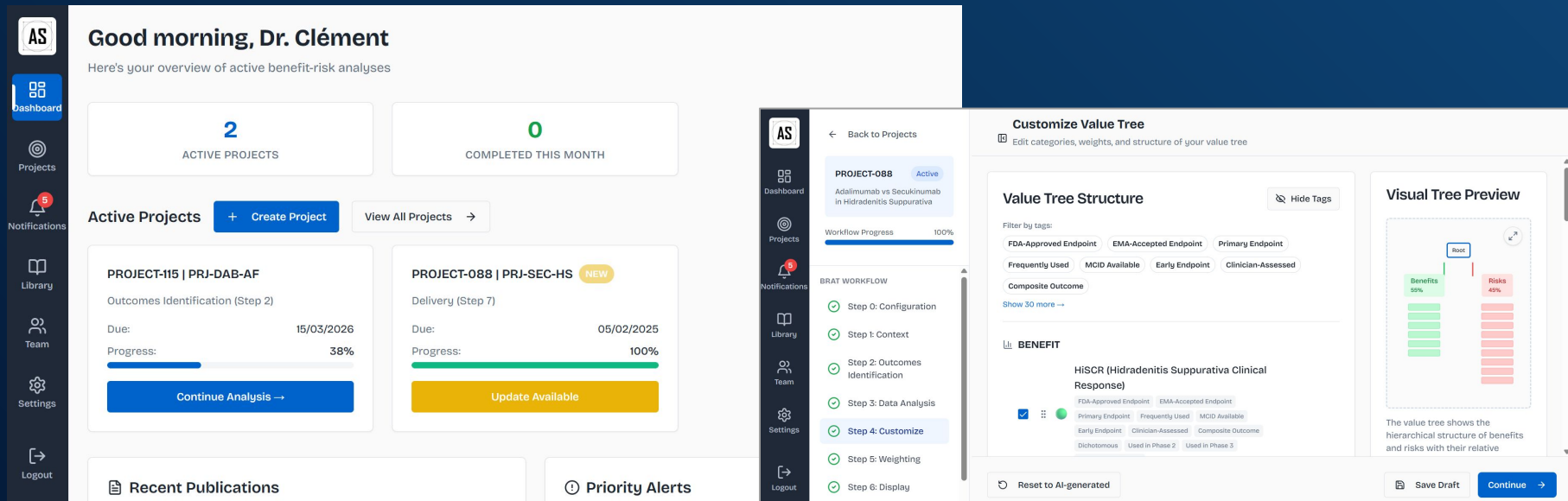
*Structured
& proven
BRAT
methodology*

Customize your BRA by having our engineers :

- Insert additional data sources
- Insert protocol design
- Select the intervention for benchmarking
- Select the appropriate outcomes
- Personalise your BR value tree and related visualisations
- Review AI general synthesis
- Generate intermediary studies

ArcaScience offers Human Oversight for Every Report

BRA Methodology Implementation by Design



Good morning, Dr. Clément
Here's your overview of active benefit-risk analyses

2 ACTIVE PROJECTS **0** COMPLETED THIS MONTH

Active Projects + Create Project View All Projects →

PROJECT-115 | PRJ-DAB-AF
Outcomes Identification (Step 2)
Due: 15/03/2026
Progress: 38%
Continue Analysis →

PROJECT-088 | PRJ-SEC-HS **NEW**
Delivery (Step 7)
Due: 05/02/2025
Progress: 100%
Update Available

Recent Publications **Priority Alerts**

Customize Value Tree
Edit categories, weights, and structure of your value tree

Value Tree Structure
Filter by tags: FDA-Approved Endpoint, EMA-Accepted Endpoint, Primary Endpoint, Frequently Used, MCID Available, Early Endpoint, Clinician-Assessed, Composite Outcome
Show 30 more →

BENEFIT
HISCR (Hidradenitis Suppurativa Clinical Response)
FDA-Approved Endpoint, EMA-Accepted Endpoint, Primary Endpoint, Frequently Used, MCID Available, Early Endpoint, Clinician-Assessed, Composite Outcome, Dichotomous, Used in Phase 2, Used in Phase 3
Reset to AI-generated

Visual Tree Preview
The value tree shows the hierarchical structure of benefits and risks with their relative

Reduce BRA Project Time by 80%¹

¹: independent BRA experts evaluation, 2025

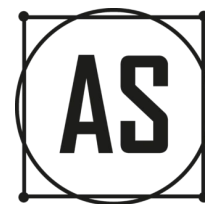
With Vs. Without BRA Platform Use

Activities	Manual BRA	Assisted BRA
Literature review	Months	Instant
Benchmarking setup	Only one (too time-consuming)	Unlimited change
Workflow management	In-person & emails	Team shares access to platform
BRA adjustment	Every parameter change generate workload	Instant impact visualisation
Outcomes	High standards require extensive manual effort	Automated excellence: Quality without the wait

The First End-to-End AI Platform for Benefit-Risk Analysis

The impact of our platform a few KPIs :

- + **Chief Medical Officer / VP Medical Affairs** (*Increase trial success rate ~6%*)
Strategic oversight of benefit-risk profile across portfolio at every stages
- + **Clinical Development Director** (*3 months/phase saved by amendment*)
Reduce amendments with trial design based on endpoint-based benefit-risk modelling
- + **Regulatory Affairs Manager** (*Benefit-Risk Assessment building time: - 80%*)
Aggregate reviewed Benefit-Risks insights to benchmark against standard of care for submissions
- + **Safety & Pharmacovigilance Director** (*2 weeks saved each month*)
Real-time Trials PICOS monitoring and case detection compatible with FDA requirement
- + **Disease Area Therapeutic Lead** (*€10k in service costs saved per area*)
Instantly reach a deep understanding of disease mechanisms and pathways
- + **Evidence Generation Manager** (*Tailored updated dataset*)
Systematic monitoring of clinical evidence



 **ArcaScience**
Seek less, discover better

Build your drugs' BRE in seconds

*Instead of months,
and recover ~90% of relevant insights
to build your clinical strategy*

Most comprehensive BRE
database in the world
unmatched, with over 100bn data points



24 small AI models
trained by clinicians



AI technology already used & documented for FDA Approval

Our BRE approach encapsulates everything about your Pathology of Interest's BR



Drug Development



Post Approval

All related Drug Benefit Risk Evaluation Predict the potential Benefits and Risks



SOC benchmarking Drug Benefit Risk Evaluation Benchmark your treatment



Competitive Landscape Explore Clinical strategies & anticipate market trends



PICOS Evidence Monitoring Monitor studies using a standard format to assess evidence



Pathology Modelisation Understand pathway & mechanisms in a targeted disease



Affiliations & Experts Mapping Find the right experts for your scientific committee



Clinical Endpoints Study Limit protocol amendments & define the best COA strategy



DDI Detector Assess all DDIs ever discussed to strengthen your drug position



SafetyShield Identify & understand every case without any time investment





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Thank you!

They trust us



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R&D Impact proven



*"As a Global R&D, I want to predict benefit-risk of my drug
from the preclinical stage down to the market
authorization (10 years)"*

**Benefit Risk Anticipation
in 2023**

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Results

For the 1st time in Sanofi's history:

successfully achieved a benefit-risk drill-down prediction based on 9 times more insights coming from internal & external sources



sanofi

Millions re-routed thanks to the anticipated confirmation of thromboembolic risks

18 months of work done in mere seconds

ArcaScience = \$100k VS CRO = \$1,2m

Unlocking success through collaboration



Goals & means:

- Out of 100 selected drugs available on the market, **find the best candidates for being repurposed to glioblastoma.**
- The first key task is to prove if there is a crossing of the BBB (Blood Brain Barrier), induced by these drugs.

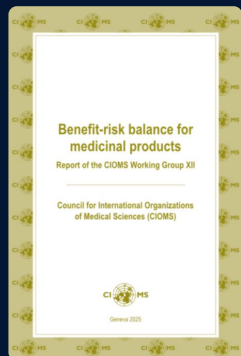
Output & take away:

- **1 dataset showcasing the safety datapoints** directly connected to the identification of the crossing of the BBB (eye deficiency, headache, earing issue, etc.)

Impact:

- **3 drugs were found** to be compatible with glioblastoma, **2 are now pushed in phase 2 of clinical trials**

2025: A Transformative Year for Benefit-Risk Analysis

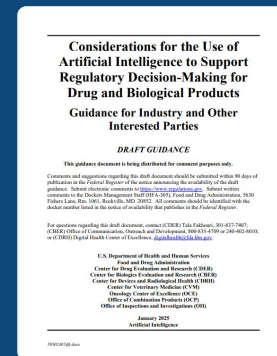


Benefit-Risk Balance for Medicinal Products - Working Group XII



Regulatory Framework Development:
CIOMS Working Group XII releases updated guidance on benefit-risk balance methodology for medicinal products

AI-Powered Transformation:
FDA publishes framework for evaluating AI model credibility and validation in drug development



Framework to Advance Credibility of AI Models Used for Drug Development